



FACULTY OF ENGINEERING TECHNOLOGY

Subject: Comm. & Data Transmission

Final Exam summer 24/25

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Duration: 90 min

Q1: choose the correct answers:

(Please write your answers in the table only; otherwise, your answers will be neglected)

(34 marks)

Question	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Answer														
Question	15	16	17											
Answer														

1. The twisted pair cable in which metal casing improves the penetration of noise or crosstalk is called

- A. insulated twisted pair cable
- ☒ B. Shielded twisted pair cable
- C. Unshielded twisted pair cable
- D. Both A and B

2. Multiplexing combines signals from several sources to achieve

- ☒ A. Data rate management
- B. Interleaving
- C. TDM efficiency
- D. bandwidth efficiency

3. The synchronous Time-Division Multiplexing (TDM) is not efficient because of

- A. higher data rate
- B. infinite frames
- ☒ C. empty slots
- D. Both A and C

4. The multiplexing technique that is used when the data rate of an input line is a multiple of the others is called

- ☒ A. multilevel multiplexing
- B. multiple-slot allocation
- C. pulse stuffing
- D. Asynchronous TDM

5. TDM is a digital multiplexing technique for combining several low-rate channels into

- A. a group of low-rate channels
- ☒ B. one high-rate one channel
- C. two high-rate channels
- D. None

6. Each TV channel has its own bandwidth of

- A. 6 Hz
- B. 6 KHz
- ☒ C. 6 MHz
- D. 60 MHz

7. In synchronous TDM, the data rate of the link is n times faster and the unit duration is:

- ☒ A. n times shorter
- B. $n+1$ times shorter
- C. $n-1$ times shorter
- D. remains the same

8. The frequency difference between the Wavelength-Division Multiplexing (WDM) and Frequency-Division Multiplexing (FDM) is

- ☒ A. very high
- B. very low
- C. zero
- D. infinity

9. In OOOVBOVB, V stands for violation, and B stands for

- A. Bit
- B. Baud
- ☒ C. Bandwidth
- D. Bipolar

10. The last step in Pulse Code Modulation (PCM) is

- A. Quantization
- B. Sampling
- ☒ C. Encoding
- D. Modulation

11. Data encoding is not possible if

- A. $2^m < L^n$
- ☒ B. $2^m > L^n$
- C. $2^m = L^n$
- D. $2^m \neq L^n$

12. In line-of-sight propagation, very high-frequency signals are transmitted in straight lines directly from the antenna to

- A. sky
- B. earth
- ☒ C. antenna
- D. planet

13. In real life, most digital signals have bandwidth

- ☒ A. Infinite
- ☐ B. Zero
- ☐ C. Constant
- ☐ D. None of the above

14. Line coding converts a sequence of

- ☐ A. Frames to digital signal
- ☒ B. bits to a digital signal
- ☐ C. Bytes to digital signal
- ☐ D. Packets of data to digital signals

15. The bandwidth is proportional to the

- ☒ A. baud rate
- ☐ B. data rate
- ☐ C. delay rate
- ☐ D. bit rate

16. Radio waves are

- ☐ A. unidirectional
- ☒ B. omnidirectional
- ☐ C. bidirectional
- ☐ D. directional

17. Optical fibers use reflection to guide light through a

- ☐ A. channel
- ☒ B. metal wire
- ☐ C. light
- ☐ D. plastic

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